

Stroke Prediction

Done by: Lab P3-1

Teng Tze Hui (2502437)

Ee Chew Fong, Olivia (2102743)

Gador Sophia Helen Santiago (2503845)

Yap Yi Hui Wynn (2501386)

[Google Colab Notebook](#)



Executive Summary & Stage 1 Baseline Recap

Bob, a 45-year-old office worker with hypertension, wants to know: "Am I at risk of having a stroke? Should I see a doctor now?"

Problem Statement:

How can health data be analyzed to provide individuals like Bob with clear, reliable insights into their stroke risk, enabling informed decisions about seeking medical advice and preventive measures?

Our priority is to **reduce False Negatives**. This is when the **model says no stroke** when Bob **has stroke**, this makes Bob **miss critical treatment** and potentially be fatal.

Stage 1 Key Findings (after SMOTE):

- Severe class imbalance: ~95% No Stroke/ ~5% Stroke
- BMI has 201 missing values (median imputation needed)
- LR strongest: highest ROC-AUC (0.837) and Balanced Accuracy (0.758)
- Preprocessing done outside pipeline (data leakage risk)

Model	F1_Score	Precision	Recall	Specificity	Accuracy	Balanced_Accuracy	ROC_AUC
Dummy Baseline	0.623165	0.892335	0.499435	0.501196	0.499435	0.485292	0.485292
Logistic Regression	0.824011	0.935783	0.760452	0.760766	0.760452	0.757934	0.836954
SVM	0.865433	0.919707	0.827119	0.848086	0.827119	0.658737	0.775754
Decision Tree	0.899913	0.900392	0.899435	0.946172	0.899435	0.524107	0.524107

Stage 2 Data Engineering

Feature Engineering:

- **New Feature: age_glucose** (age x avg_glucose)
- Captures the **interaction** effect of **age** and **glucose** level for stroke risk
- Added via separate function **to ensure consistent transformation**
- No missing value in **age** or **avg_glucose_level**

Binary Encoding

- In **Stage1: Manual .map()**, outside pipeline
- In **Stage 2: OrdinalEncoder**, inside pipeline.
 - Columns: gender, ever_married, Residence_Type
 - **Fits only on training set, this eliminates data leakage risk**

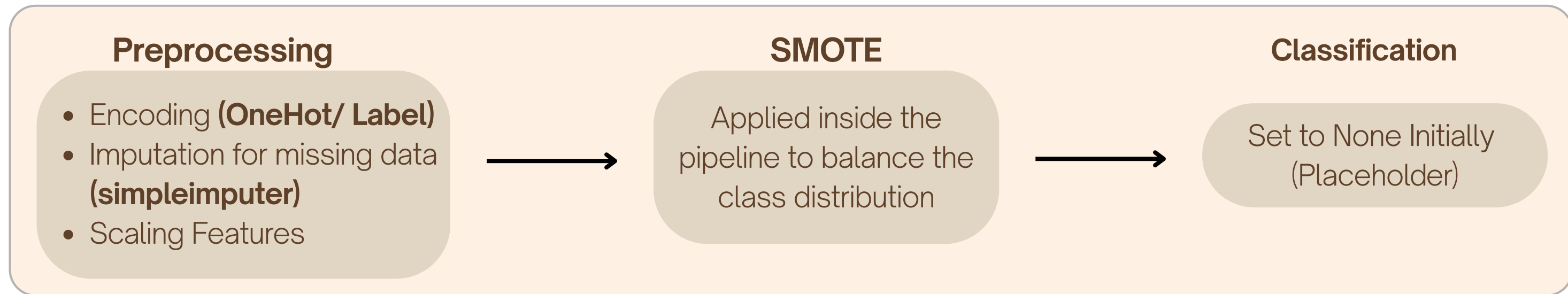
Nominal Encoding

- In **Stage1: OneHotEncoder**, outside pipeline
- In **Stage 2: OneHotEncoder**, inside pipeline
 - Columns: work_type, smoking_status handle_unknown = "ignore" **for robust unseen categories**

Imputation + Scaling: Chained in Pipeline

- In **Stage1: SimpleImputer + StandardScaler** called manually
- In **Stage 2: Nested pipeline inside Column Transformer**
 - **Median imputation** for BMI, StandardScaler **skipped** for Random Forest (Pass through)

Pipeline Architecture



1) **Preprocessing**: Handles encoding, imputing missing values, and scaling features.

- Safely fit only on training data to prevent leakage.

2) **SMOTE**: Synthetic Minority Over-sampling Technique used to balance class distribution

- Within the pipeline to avoid leakage.
- SMOTE is applied via `imblearn.Pipeline` to ensure oversampling occurs only on training folds, not validation folds.

3) **Classification**: Placeholder for classifier, set to None initially to ensure isolation from preprocessing steps.

All preprocessing steps — median imputation, ordinal/one-hot encoding, and `StandardScaler` — were strictly encapsulated within `sklearn`.

All steps are chained inside Pipeline objects, ensuring zero data leakage during cross-validation. Transformations were fitted only on training folds and applied to validation folds, never the reverse.



Model Selection

Model Selection Justification:

- Logistic Regression provided a strong baseline in Stage 1 with high ROC AUC and Recall, while SVM achieved slightly higher F1 and excellent Specificity; Decision Tree was biased toward the majority class and replaced by **Random Forest** (tree-based ensemble).
- In Stage 2, **Logistic Regression** (linear model) was retained for its baseline performance, Random Forest was added to reduce bias and capture non-linear patterns through ensemble learning, and the **Stacked LR + SVM** (linear + distance-based) model was included to leverage complementary strengths, aiming to maximize F1 and Recall for stroke prediction, with Logistic Regression as the final estimator for stable, interpretable probabilistic predictions.

Cross-validation setup:

- 5-fold Stratified K-Fold to maintain class balance.
 - shuffle=True, random_state=42 for reproducibility.
- Ensures reliable performance estimates and captures variability across folds.

	Model	F1_Score	Precision	Recall	Specificity	Balanced_Accuracy	ROC_AUC	F1_Score_STD	Balanced_Accuracy_STD	ROC_AUC_STD
0	Logistic Regression	0.820070	0.929757	0.756017	0.760119	0.723457	0.808628	0.012489	0.030184	0.011366
2	Stacking (LR + SVM)	0.873267	0.910839	0.843944	0.874221	0.604097	0.680366	0.010174	0.031697	0.069617
1	Random Forest	0.915591	0.901506	0.935258	0.988621	0.512067	0.756641	0.002970	0.006713	0.026466

Evaluation:

- Logistic Regression: **Balanced** performance with **high Balanced Accuracy** (0.723) and **stable ROC-AUC** (0.809 ± 0.011). **Reliable** across folds.
- Random Forest: **High Recall** (0.935), but **low Balanced Accuracy** (0.512) and **moderate ROC-AUC** (0.757 ± 0.026). Performance **biased** toward majority class.
- Stacked LR + SVM: **Moderate** Balanced Accuracy (0.604) and ROC-AUC (0.680 ± 0.070). ROC-AUC variability is high (STD = 0.070), indicating **less stability**.
- Note: Metrics reference Experiment 2 because it represents the first reliable setup with minority-class improvement using BorderlineSMOTE.



Justification of Champion Model

Selection Criteria:

- Balanced Accuracy: Accounts for class imbalance by averaging recall across classes.
- ROC AUC: Evaluates class separation across thresholds, independent of decision threshold.
- Recall: Prioritizes detection of actual stroke cases (minority class).
- Clinical Relevance: maximise true positive (classified stroke & has stroke) and minimise false negatives (classified as no stroke but has stroke)
- Stability: Ensures reliable performance across folds

Random Forest

- High Recall (0.935) is misleading — almost all predictions are “No Stroke,” resulting in **extremely low True Positives** (7).
- Low Balanced Accuracy (0.512) indicates **poor minority-class handling**.
- Despite strong aggregate metrics, **fails to detect actual strokes** making it clinically ineffective.

Stacking (LR + SVM)

- Moderate Recall (0.844) and Balanced Accuracy (0.604), but high ROC-AUC variability (STD = 0.070) → **unreliable** class separation across folds.
- Detects fewer strokes than LR (66 TP) → **less clinically meaningful**.

Logistic Regression

- **Highest** Balanced Accuracy (0.723) → most effective at handling class imbalance and maintaining overall predictive performance.
- **Stable** ROC-AUC (0.809 ± 0.011) → consistently separates stroke vs. non-stroke across folds.
- **Highest** True Positives (136) → maximizes detection of clinically relevant stroke cases.
- Slightly higher False Positive rate is acceptable because it is **safer** than missing strokes in the clinical context.

Champion Model: Logistic Regression

- Best overall balance between class separation, minority class performance, stability, and clinically meaningful stroke prediction.

Criterion	Logistic Regression	Random Forest	Stacking (LR + SVM)
Balanced Accuracy	0.723457	0.512067	0.604097
ROC-AUC	0.8086 ± 0.011	0.7566 ± 0.026	0.6804 ± 0.070
Recall	0.756	0.935*	0.844
True Positives	136	7	66
False Negatives	62	191	132
Stability (STD)	Low	Medium	High

Controlled Ablation & Tuning Log

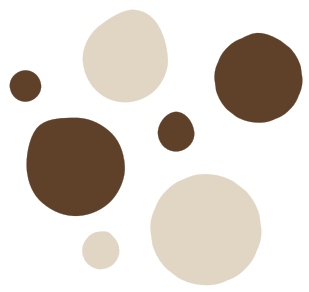
Exp	Hypothesis	Controlled Change	CV Balanced Accuracy (Mean ± Std)	CV ROC-AUC (Mean ± Std)	FN Count*
1	Baseline LR	Baseline Logistic Regression + SMOTE	0.7358 ± 0.0181	0.8086 ± 0.0145	49
2	BorderlineSMOTE improves minority detection over SMOTE.	Replaced SMOTE with BorderlineSMOTE	0.7235 ± 0.0302	0.8086 ± 0.0114	62
3	Adding <code>age x avg_glucose_level</code> improves results.	Added engineered feature <code>age_glucose</code>	0.7235 ± 0.0302	0.8086 ± 0.0114	62
4	Hypertuning improves Logistic Regression generalisation.	RandomizedSearchCV on <code>C</code> , <code>penalty</code> , <code>solver</code>	0.7323 ± 0.0334	0.8092 ± 0.0152	56

Four controlled experiments were conducted exclusively on the champion model:

- Experiment 1: **Refrence / Strongest overall** on the primary metric
- Experiment 2: **Mixed impact** - FNs increased and Balanced Accuracy decreased.
- Experiment 3: **Dropped** - No meaningful improvement over Experiment 2
- Experiment 4: **Marginal gain** over Experiment 2, but still below Experiment 1 on Balanced Accurcacy

Thus, after the model champion selection, and recognizing the results from all experiments, we decided to revert to SMOTE as it still provides the best performance for stroke detection.

As a result, the final champion model is the **Logistic Regression model with SMOTE**, as it achieved the highest Balanced Accuracy and the most stable results.



Final Model Stability Metrics

Final Model Selected: Logistic Regression + SMOTE
Evaluation: 5-fold Stratified Cross-Validation

Balanced Accuracy	0.7358 ± 0.0181
ROC - AUC	0.8086 ± 0.0145
F1 Score	0.7960 ± 0.0109

- Low standard deviation → stable performance across folds
- Strong ROC-AUC → good class separation
- Balanced Accuracy → handles class imbalance well

Mechanical Failure Analysis

... === FALSE NEGATIVES (Missed Strokes) ===

	age	avg_glucose_level	bmi	hypertension	heart_disease	ever_married	actual	predicted	probability
33	32.0	76.13	29.9	0	0	Yes	1	0	0.063265
1072	43.0	143.43	45.9	0	0	Yes	1	0	0.089923
642	39.0	83.24	26.3	0	0	Yes	1	0	0.103078
1947	42.0	83.41	25.4	0	0	Yes	1	0	0.103683
3136	39.0	97.76	29.6	0	0	Yes	1	0	0.116171
2869	45.0	64.14	29.4	0	0	Yes	1	0	0.121740
2342	48.0	84.20	29.7	0	0	No	1	0	0.126564
2866	38.0	82.28	24.0	0	0	No	1	0	0.146578
3235	38.0	101.45	NaN	0	0	Yes	1	0	0.175405
1025	46.0	78.18	30.8	0	0	Yes	1	0	0.192179

Row-level inspection revealed that the model’s most critical failures occur in patients who resemble Bob in age but lack his hypertension. These are typically younger patients (aged 32–48) with normal glucose levels and no classic risk indicators, yet they are assigned extremely low stroke probabilities, in some cases as low as 0.006, despite being confirmed stroke cases.

This suggests that the model over-relies on dominant features such as age and glucose, creating a blind spot for atypical risk profiles. While Bob’s hypertension makes him a stronger candidate for detection, patients without such indicators are at risk of being silently misclassified, which represents the most dangerous failure in a healthcare context.



Proposed Fixes for Observed Failures

Error	Scenario for Bob	Real-world Cost
False Negative	Model says no stroke, Bob has stroke	Misses treatment → potentially fatal
False Positive	Model says stroke, Bob has no stroke	Unnecessary consultation → stressful but safe

By default, classifiers predict stroke when the probability is greater than or equal to 0.5. However, based on our Mechanical Failure Analysis, false negatives—where actual stroke cases are missed—are significantly more harmful than false positives. A missed stroke means that the patient receives no warning and may skip critical early intervention, potentially leading to severe or fatal outcomes. In contrast, a false positive only results in an unnecessary medical consultation, which is a safer alternative. Therefore, we search for an optimal decision threshold that maximises Balanced Accuracy, our primary evaluation metric, while better reflecting the real-world cost of prediction errors. This ensures that the model prioritises reducing missed stroke cases and aligns with the objective of safeguarding patients like Bob.

As a result, the threshold was reduced from 0.5 to 0.453 to improve the model’s sensitivity to stroke cases.

Best Threshold: 0.4530

Best Balanced Accuracy: 0.7498



Business Decision for Threshold Shift

	Threshold 0.5 (Default)	Threshold 0.453 (Optimal)
Balanced Accuracy	0.7258	0.7304
Recall (TPR)	0.7525	0.8081
Specificity	0.7191	0.6864
Missed Strokes (FN)	49	38
False Alarms (FP)	938	1047

The business decision centres on adjusting the classification threshold to better align the model's behaviour with real-world clinical priorities. While the default threshold of 0.5 provides a balanced trade-off between precision and recall, it results in a higher number of missed stroke cases, which is unacceptable in a healthcare context. By lowering the threshold to 0.453, the model becomes more sensitive in identifying potential stroke cases, increasing recall from 0.7525 to 0.8081 and reducing false negatives from 49 to 38. This means that more patients at risk of stroke are correctly flagged for further medical attention. Although this adjustment leads to an increase in false positives, this trade-off is considered acceptable because the consequences of a false positive—an unnecessary consultation—are significantly less severe than those of a false negative, which may result in missed treatment and potentially fatal outcomes. Therefore, the final policy deliberately prioritises patient safety over predictive precision, ensuring that the model errs on the side of caution and minimises the risk of undetected stroke cases in practical deployment.

Individual Contributions

Teng Tze Hui (2502437)

- Conducted hyperparameter tuning (Exp 4) using RandomisedSearchCV for regularisation strengths and solver configurations.
- Performed final model evaluation on the held-out test set, documenting metrics like ROC AUC, Balanced Accuracy, and F1 Score.
- Adjusted thresholds to optimise model performance based on evaluation results.

Ee Chew Fong, Olivia (2102743)

- Researched BorderlineSMOTE as an alternative oversampling strategy and implemented it within the sklearn pipeline
- Conducted controlled comparison against standard SMOTE (Exp 2) across recall, balanced accuracy, and false negative rate
- Documented findings and contributed to the ablation experiment analysis

Yap Yi Hui Wynn (2501386)

- Refactored BMI imputation and preprocessing transforms into a fully encapsulated sklearn Pipeline, ensuring zero data leakage
- Implemented feature engineering experiments (Exp 3) to test whether derived features could improve model performance
- Validated pipeline integrity across all CV splits, confirming all transformers were fitted on training data only

Gador Sophia Helen Santiago (2503845)

- Implemented and evaluated multiple algorithmic families — Logistic Regression, Random Forest, and Stacking ensemble
- Built the Stage 1 baseline model (Exp 1) and established the initial evaluation framework using stratified cross-validation
- Compared model performance across balanced accuracy, ROC AUC, and F1 to support champion model selection

References

- <https://machinelearningmastery.com/mastering-the-art-of-hyperparameter-tuning-tips-tricks-and-tools/>
- <https://www.geeksforgeeks.org/machine-learning/hyperparameter-tuning/>
- <https://medium.com/@nirajan.acharya777/do-you-really-think-hypertuning-improve-models-accuracy-cfc59e41cc10>
- <https://www.analyticsvidhya.com/blog/2022/02/a-comprehensive-guide-on-hyperparameter-tuning-and-its-techniques/>
- <https://www.geeksforgeeks.org/machine-learning/hyperparameter-optimization-based-on-bayesian-optimization/>
- https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html
- https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.RandomizedSearchCV.html
- <https://machinelearningmastery.com/hyperparameters-for-classification-machine-learning-algorithms/>
- <https://codesignal.com/learn/courses/intro-to-model-optimization-in-machine-learning/lessons/hyperparameter-tuning-in-logistic-regressions>
- <https://medium.com/codex/do-i-need-to-tune-logistic-regression-hyperparameters-1cb2b81fca69>
- <https://www.kaggle.com/code/funxexcel/p2-logistic-regression-hyperparameter-tuning>
- <https://levelup.gitconnected.com/a-comprehensive-analysis-of-hyperparameter-optimization-in-logistic-regression-models-521564c1bfc0>
- <https://www.digitalocean.com/community/tutorials/logistic-regression-with-scikit-learn>





Thank You

